相机字段标签与格式化参考

本参考按当前 CameraDebugFormatter 和 CameraDebugSnapshot 整理字段身份、显示标签、值表达式与格式化调用。用于开发回查，不作为固件协议物理单位或完整枚举规范。

版本范围 p/caijiamin/feat_debug_2th 9010928ca51c | 当前实现与明确标注的待办

阅读规则

字段 ID 属于详情展示层稳定身份，值表达式指向快照属性；它们不自动等于新 Registry 的 sourceId 与 fieldId。字段存在于详情 Formatter，也不表示已在通用浮窗注册。

格式化为整数的温度、电压、状态不补写未核实单位。未知版本和无效 Dev 数据按 Mapper 规则处理。名称单独查询、身份尾号与完整复制文本的区别详见相机身份说明。

四组状态

媒体组包含直接模式、开机模式和录像状态；设备组包含系统、蓝牙、版本、电池和连接；能力组当前两项；路由组当前四项。正文附上从源码核对提取的字段表和快照模型成员表。

媒体显示字段

显示 ID	显示标签	值表达式
media.main_work_mode	直接字段 / 主工作模式	formatInt(status.mainWorkMode)
media.sub_work_mode	直接字段 / 子工作模式	formatInt(status.subWorkMode)
media.media_status	直接字段 / 媒体状态	formatInt(status.mediaStatus)
media.boot.valid	BootUp / 数据有效	formatBoolean(status.bootUpWorkMode.valid)
media.boot.main_work_mode	BootUp / 主工作模式	formatInt(status.bootUpWorkMode.mainWorkMode)
media.boot.sub_work_mode	BootUp / 子工作模式	formatInt(status.bootUpWorkMode.subWorkMode)
media.boot.action	BootUp / 动作	formatInt(status.bootUpWorkMode.action)
media.record.recording	Record / 正在录像	formatBoolean(status.recordStatus.recording)
media.record.storing	Record / 正在保存	formatBoolean(status.recordStatus.storing)
media.record.duration	Record / 录像时长	formatDuration(status.recordStatus.durationSeconds)
media.record.deleting	Record / 正在删除	formatBoolean(status.recordStatus.deleting)
media.record.state	Record / 原始状态	formatRecordState(status.recordStatus.state)

系统显示字段

显示 ID	显示标签	值表达式
dev.system.power_value	System / 电源值	formatInt(status.powerValue)
dev.system.cpu_temperature	System / CPU 温度	formatInt(status.cpuTemperature)
dev.system.temperature_state	System / 温度状态	formatInt(status.temperatureState)
dev.system.usb_setting_mode	System / USB 设置模式	formatInt(status.usbSettingMode)
dev.system.buzzer_enabled	System / 蜂鸣器启用	formatBoolean(status.buzzerEnabled)
dev.system.led_brightness	System / LED 亮度	formatInt(status.ledBrightness)
dev.system.led_enabled	System / LED 启用	formatBoolean(status.ledEnabled)
dev.system.auto_suspend_time	System / 自动休眠时间	formatInt(status.autoSuspendTime)
dev.system.a_board_pcb_check	System / A 板 PCB 检测	formatBoolean(status.aBoardPcbCheckFlag)
dev.system.sensor_pcb_check	System / Sensor PCB 检测	formatBoolean(status.sensorPcbCheckFlag)
dev.system.device_model	System / 相机型号	formatNullableText(status.deviceModel)
dev.system.device_sn	System / 相机 SN	formatNullableText(status.deviceSn)
dev.system.battery_temperature_state	System / 电池温度状态	formatInt(status.batteryTemperatureState)
dev.system.battery_capacity_state	System / 电池容量状态	formatInt(status.batteryCapacityState)
dev.system.adapter_power_state	System / 适配器电源状态	formatInt(status.adapterPowerState)
dev.system.ready	System / 设备就绪	formatBoolean(status.ready)

蓝牙显示字段

显示 ID	显示标签	值表达式
dev.bluetooth.state	Bluetooth / 状态	formatInt(status.bluetoothStatus.state)
dev.bluetooth.mac_address	Bluetooth / MAC 地址	formatMacAddress(status.bluetoothStatus.macAddress)

版本显示字段

显示 ID	显示标签	值表达式
dev.version.packet_version	Version / 数据包版本	formatVersionRaw(status.packetVersion)
dev.version.fallback_version	Version / 兼容版本	formatVersionRaw(status.fallbackVersion)
dev.version.product_type	Version / 产品类型	formatNullableText(status.productType)
dev.version.firmware_type	Version / 固件类型	formatNullableText(status.firmwareType)
dev.version.platform_type	Version / 平台类型	formatNullableText(status.platformType)
dev.version.device_type	Version / 设备类型	formatNullableText(status.deviceType)
dev.version.camera_hardware	Version / 相机硬件版本	formatVersionRaw(status.cameraHardwareVersion)
dev.version.camera_software	Version / 相机软件版本	formatVersionRaw(status.cameraSoftwareVersion)
dev.version.lens_hardware	Version / 镜头硬件版本	formatVersionRaw(status.lensHardwareVersion)
dev.version.lens_software	Version / 镜头软件版本	formatVersionRaw(status.lensSoftwareVersion)
dev.version.tof_software	Version / TOF 软件版本	formatVersionRaw(status.tofSoftwareVersion)
dev.version.battery_hardware	Version / 电池硬件版本	formatVersionRaw(status.batteryHardwareVersion)
dev.version.battery_software	Version / 电池软件版本	formatVersionRaw(status.batterySoftwareVersion)
dev.version.ai_software	Version / AI 软件版本	formatVersionRaw(status.aiSoftwareVersion)
dev.version.gimbal_hardware	Version / 云台硬件版本	formatVersionRaw(status.gimbalHardwareVersion)
dev.version.gimbal_software	Version / 云台软件版本	formatVersionRaw(status.gimbalSoftwareVersion)
dev.version.bluetooth_software	Version / 蓝牙软件版本	formatVersionRaw(status.bluetoothSoftwareVersion)
dev.version.system_manager_software	Version / 系统管理软件版本	formatVersionRaw(status.systemManagerSoftwareVersion)
dev.version.remote_hardware	Version / 遥控器硬件版本	formatVersionRaw(status.remoteHardwareVersion)
dev.version.remote_software	Version / 遥控器软件版本	formatVersionRaw(status.remoteSoftwareVersion)
dev.version.remote_sub_mcu	Version / 遥控器子 MCU 版本	formatVersionRaw(status.remoteSubMcuVersion)

显示 ID	显示标签	值表达式
dev.version.swivel_hardware	Version / 转轴硬件版本	formatVersionRaw(status.swivelHardwareVersion)
dev.version.swivel_software	Version / 转轴软件版本	formatVersionRaw(status.swivelSoftwareVersion)

电池显示字段

显示 ID	显示标签	值表达式
dev.battery.status	Battery / 状态	formatInt(status.status)
dev.battery.capacity	Battery / 电量	formatPercent(status.capacityPercent)
dev.battery.charging	Battery / 正在充电	formatBoolean(status.charging)
dev.battery.differential_voltage	Battery / 压差	formatInt(status.differentialVoltage)
dev.battery.pcb_check	Battery / PCB 检测	formatBoolean(status.pcbCheckFlag)
dev.battery.voltage	Battery / 电压	formatInt(status.voltage)
dev.battery.temperature	Battery / 温度	formatInt(status.temperature)
dev.battery.power_action	Battery / 电源动作	formatInt(status.powerAction)
dev.battery.adapter_plugged	Battery / 适配器已连接	formatBoolean(status.adapterPlugged)
dev.battery.light_operation	Battery / 灯光操作	formatInt(status.lightOperation)
dev.battery.light_slider	Battery / 灯光亮度	formatInt(status.lightSlider)

连接显示字段

显示 ID	显示标签	值表达式
dev.connection.ai_online	Connect / AI 在线	formatBoolean(status.aiOnline)
dev.connection.gimbal_online	Connect / 云台在线	formatBoolean(status.gimbalOnline)
dev.connection.battery_online	Connect / 电池在线	formatBoolean(status.batteryOnline)
dev.connection.lens_online	Connect / 镜头在线	formatBoolean(status.lensOnline)
dev.connection.tof_online	Connect / TOF 在线	formatBoolean(status.tofOnline)
dev.connection.bluetooth_online	Connect / 蓝牙在线	formatBoolean(status.bluetoothOnline)
dev.connection.usb_wifi_attached	Connect / USB Wi-Fi 已连接	formatBoolean(status.usbWifiAttached)
dev.connection.poe_attached	Connect / PoE 已连接	formatBoolean(status.poeAttached)
dev.connection.swivel_base_attached	Connect / 转轴底座已连接	formatBoolean(status.swivelBaseAttached)
dev.connection.uwb_attached	Connect / UWB 已连接	formatBoolean(status.uwbAttached)
dev.connection.audio_35mm_attached	Connect / 3.5mm 音频已连接	formatBoolean(status.audio35mmAttached)
dev.connection.audio_35mm_line_attached	Connect / 3.5mm Line 已连接	formatBoolean(status.audio35mmLineAttached)
dev.connection.usb_c_audio_attached	Connect / USB-C 音频已连接	formatBoolean(status.usbCAudioAttached)
dev.connection.voice_device_attached	Connect / 语音设备已连接	formatBoolean(status.voiceDeviceAttached)
dev.connection.hdmi_attached	Connect / HDMI 已连接	formatBoolean(status.hdmiAttached)
dev.connection.px30_attached	Connect / PX30 已连接	formatBoolean(status.px30Attached)
dev.connection.sd_card_inserted	Connect / SD 卡已插入	formatBoolean(status.sdCardInserted)
dev.connection.remote_attached	Connect / 遥控器已连接	formatBoolean(status.remoteAttached)
dev.connection.sensor_error	Connect / Sensor 异常	formatBoolean(status.sensorError)

显示 ID	显示标签	值表达式
dev.connection.media_error	Connect / 媒体异常	formatBoolean(status.mediaError)

能力显示字段

显示 ID	显示标签	值表达式
capability.support_broadcasting	支持直播	formatBoolean(status.supportBroadcasting)
capability.support_usb_mic	支持 USB 麦克风	formatBoolean(status.supportUsbMic)

路由显示字段

显示 ID	显示标签	值表达式
router.total_count	连接总数	formatInt(status.totalCount)
router.app_count	App 连接数	formatInt(status.appCount)
router.monitor_count	Monitor 连接数	formatInt(status.monitorCount)
router.pc_count	PC 连接数	formatInt(status.pcCount)

只读快照成员

模型	成员	Kotlin 类型
CameraDebugSnapshot	uniKey	String
CameraDebugSnapshot	receivedAtMillis	Long
CameraDebugSnapshot	mediaStatus	CameraMediaDebugSnapshot?
CameraDebugSnapshot	devStatus	CameraDevDebugSnapshot?
CameraDebugSnapshot	capability	CameraCapabilityDebugSnapshot?
CameraDebugSnapshot	routerStatus	CameraRouterDebugSnapshot?
CameraDebugSnapshot	deviceName	String?
CameraMediaDebugSnapshot	mainWorkMode	Int
CameraMediaDebugSnapshot	subWorkMode	Int
CameraMediaDebugSnapshot	mediaStatus	Int
CameraMediaDebugSnapshot	bootUpWorkMode	CameraBootUpWorkModeDebugSnapshot
CameraMediaDebugSnapshot	recordStatus	CameraRecordDebugSnapshot
CameraBootUpWorkModeDebugSnapshot	valid	Boolean
CameraBootUpWorkModeDebugSnapshot	mainWorkMode	Int
CameraBootUpWorkModeDebugSnapshot	subWorkMode	Int
CameraBootUpWorkModeDebugSnapshot	action	Int
CameraRecordDebugSnapshot	recording	Boolean
CameraRecordDebugSnapshot	storing	Boolean
CameraRecordDebugSnapshot	durationSeconds	Int
CameraRecordDebugSnapshot	deleting	Boolean
CameraRecordDebugSnapshot	state	Int

模型	成员	Kotlin 类型
CameraDevDebugSnapshot	systemStatus	CameraSystemDebugSnapshot
CameraDevDebugSnapshot	versionStatus	CameraVersionDebugSnapshot
CameraDevDebugSnapshot	batteryStatus	CameraBatteryDebugSnapshot
CameraDevDebugSnapshot	connectionStatus	CameraDeviceConnectionDebugSnapshot
CameraSystemDebugSnapshot	powerValue	Int
CameraSystemDebugSnapshot	cpuTemperature	Int
CameraSystemDebugSnapshot	temperatureState	Int
CameraSystemDebugSnapshot	usbSettingMode	Int
CameraSystemDebugSnapshot	buzzerEnabled	Boolean
CameraSystemDebugSnapshot	ledBrightness	Int
CameraSystemDebugSnapshot	ledEnabled	Boolean
CameraSystemDebugSnapshot	autoSuspendTime	Int
CameraSystemDebugSnapshot	aBoardPcbCheckFlag	Boolean
CameraSystemDebugSnapshot	sensorPcbCheckFlag	Boolean
CameraSystemDebugSnapshot	deviceModel	String?
CameraSystemDebugSnapshot	deviceSn	String?
CameraSystemDebugSnapshot	batteryTemperatureState	Int
CameraSystemDebugSnapshot	batteryCapacityState	Int
CameraSystemDebugSnapshot	adapterPowerState	Int
CameraSystemDebugSnapshot	ready	Boolean

模型	成员	Kotlin 类型
CameraSystemDebugSnapshot	bluetoothStatus	CameraBluetoothDebugSnapshot
CameraBluetoothDebugSnapshot	state	Int
CameraBluetoothDebugSnapshot	macAddress	List<Byte>?
CameraVersionDebugSnapshot	packetVersion	Int?
CameraVersionDebugSnapshot	fallbackVersion	Int?
CameraVersionDebugSnapshot	productType	String?
CameraVersionDebugSnapshot	firmwareType	String?
CameraVersionDebugSnapshot	platformType	String?
CameraVersionDebugSnapshot	deviceType	String?
CameraVersionDebugSnapshot	cameraHardwareVersion	Int?
CameraVersionDebugSnapshot	cameraSoftwareVersion	Int?
CameraVersionDebugSnapshot	lensHardwareVersion	Int?
CameraVersionDebugSnapshot	lensSoftwareVersion	Int?
CameraVersionDebugSnapshot	tofSoftwareVersion	Int?
CameraVersionDebugSnapshot	batteryHardwareVersion	Int?
CameraVersionDebugSnapshot	batterySoftwareVersion	Int?
CameraVersionDebugSnapshot	aiSoftwareVersion	Int?
CameraVersionDebugSnapshot	gimbalHardwareVersion	Int?
CameraVersionDebugSnapshot	gimbalSoftwareVersion	Int?
CameraVersionDebugSnapshot	bluetoothSoftwareVersion	Int?

模型	成员	Kotlin 类型
CameraVersionDebugSnapshot	systemManagerSoftwareVersion	Int?
CameraVersionDebugSnapshot	remoteHardwareVersion	Int?
CameraVersionDebugSnapshot	remoteSoftwareVersion	Int?
CameraVersionDebugSnapshot	remoteSubMcuVersion	Int?
CameraVersionDebugSnapshot	swivelHardwareVersion	Int?
CameraVersionDebugSnapshot	swivelSoftwareVersion	Int?
CameraBatteryDebugSnapshot	status	Int
CameraBatteryDebugSnapshot	capacityPercent	Int
CameraBatteryDebugSnapshot	charging	Boolean
CameraBatteryDebugSnapshot	differentialVoltage	Int
CameraBatteryDebugSnapshot	pcbCheckFlag	Boolean
CameraBatteryDebugSnapshot	voltage	Int
CameraBatteryDebugSnapshot	temperature	Int
CameraBatteryDebugSnapshot	powerAction	Int
CameraBatteryDebugSnapshot	adapterPlugged	Boolean
CameraBatteryDebugSnapshot	lightOperation	Int
CameraBatteryDebugSnapshot	lightSlider	Int
CameraDeviceConnectionDebugSnapshot	aiOnline	Boolean
CameraDeviceConnectionDebugSnapshot	gimbalOnline	Boolean
CameraDeviceConnectionDebugSnapshot	batteryOnline	Boolean

模型	成员	Kotlin 类型
CameraDeviceConnectionDebugSnapshot	lensOnline	Boolean
CameraDeviceConnectionDebugSnapshot	tofOnline	Boolean
CameraDeviceConnectionDebugSnapshot	bluetoothOnline	Boolean
CameraDeviceConnectionDebugSnapshot	usbWifiAttached	Boolean
CameraDeviceConnectionDebugSnapshot	poeAttached	Boolean
CameraDeviceConnectionDebugSnapshot	swivelBaseAttached	Boolean
CameraDeviceConnectionDebugSnapshot	uwbAttached	Boolean
CameraDeviceConnectionDebugSnapshot	audio35mmAttached	Boolean
CameraDeviceConnectionDebugSnapshot	audio35mmLineAttached	Boolean
CameraDeviceConnectionDebugSnapshot	usbCAudioAttached	Boolean
CameraDeviceConnectionDebugSnapshot	hdmiAttached	Boolean
CameraDeviceConnectionDebugSnapshot	px30Attached	Boolean
CameraDeviceConnectionDebugSnapshot	sdCardInserted	Boolean
CameraDeviceConnectionDebugSnapshot	remoteAttached	Boolean
CameraDeviceConnectionDebugSnapshot	sensorError	Boolean
CameraDeviceConnectionDebugSnapshot	mediaError	Boolean
CameraDeviceConnectionDebugSnapshot	voiceDeviceAttached	Boolean
CameraCapabilityDebugSnapshot	supportBroadcasting	Boolean
CameraCapabilityDebugSnapshot	supportUsbMic	Boolean
CameraRouterDebugSnapshot	totalCount	Int

模型	成员	Kotlin 类型
CameraRouterDebugSnapshot	appCount	Int
CameraRouterDebugSnapshot	monitorCount	Int
CameraRouterDebugSnapshot	pcCount	Int

格式化与缺失规则

函数	实际规则
formatBoolean	true 显示是 false 显示否
formatInt	原始整数转字符串 不添加物理单位
formatDuration	非负数添加秒 负数显示 --
formatPercent	非负数添加 % 负数显示 -- 当前不限制大于 100
formatRecordState	0 Idle 1 Starting 2 Running 3 Stopping 其他保留未知原值
formatVersionRaw	null 显示 -- 其他十进制加八位十六进制
formatMacAddress	空列表显示 -- 每字节与 0xFF 后按两位十六进制连接
formatIdentityTail	SN 优先 无 SN 时 uniKey 取后四位

源码依据与版本回查

统一基线提交 9010928ca51c6e766f75414c49c748338e0b7ecb。以下摘要为文件 SHA256 前 16 位，用于核对资料整理时的源码版本。

business/debug/debug_impl/src/main/java/com/obsbot/live/business/debug/ui/cmera/CameraDebugFormatter.kt

摘要 7797af3e102470cf；主要符号 CameraDebugFormatter、formatGroups、formatCameraName、formatCameraModel、formatIdentityTail、formatReceivedAt、formatCopyText、formatMediaStatus、formatDevStatus、MutableList、formatCapability、formatRouterStatus

business/debug/debug_impl/src/main/java/com/obsbot/live/business/debug/CameraDebugSnapshot.kt

摘要 0b5b57ba5218a380；主要符号 CameraDebugSnapshot、CameraMediaDebugSnapshot、CameraBootUpWorkModeDebugSnapshot、CameraRecordDebugSnapshot、CameraDevDebugSnapshot、CameraSystemDebugSnapshot、CameraBluetoothDebugSnapshot、CameraVersionDebugSnapshot、CameraBatteryDebugSnapshot、CameraDeviceConnectionDebugSnapshot、CameraCapabilityDebugSnapshot、CameraRouterDebugSnapshot

business/debug/debug_impl/src/main/java/com/obsbot/live/business/debug/CameraDebugSnapshotMapper.kt

摘要 df922c3f8bc4aaa7；主要符号 CameraDebugSnapshotMapper、map、mapMediaStatus、mapDevStatus、mapCapability、mapRouterStatus、Int.toVersionOrNull、String